



WA-7436

**Integrated M. C. A. (Sem. IV)
Examination, June-2024**

IMCA - 401

Operating System

Time Allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

Note : This question paper is divided into three sections as under :

Section-A**Max. Marks-10**

This section shall have 01 compulsory question comprising 10 questions (Maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

Section-B**Max. Marks-25**

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section-C**Max. Marks-35**

Will carry 35 marks with five long answer type questions comprising one compulsory question (question no. 01) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

SECTION - A

- 1
 - (i) What is Parallel systems ?
 - (ii) Explain the term scheduler.
 - (iii) Define producer - consumer problem of synchronization.
 - (iv) Define semaphore in critical region.
 - (v) Write two advantages of virtual memory.
 - (vi) What is the basic function of paging ?
 - (vii) Explain storage hierarchy of a computer system.
 - (viii) Differentiate between sequential and index sequential access.
 - (ix) What is the need of Authentication ?
 - (x) Define File Attributes.

SECTION - B**UNIT - I**

- 2 Describe the process state with the help of state diagram. Define PCB and its role in the context switching of the process.

OR

Define the following terms :

- (1) Turnaround time
- (2) Waiting time
- (3) Throughput
- (4) CPU utilization
- (5) Response time

UNIT - II

- 3 Write short notes on :
- (i) What is critical section ?
 - (ii) Explain the techniques to prevent the dead lock.

OR

What is synchronization ? Discuss any two classical problems of synchronization.

UNIT - III

- 4 Explain the concept of demand paging with neat diagram.

OR

Given memory partition of 100 KB, 500 KB, 200 KB and 800 KB. Show with neat sketch how would each of the first fit, best fit and worst fit algorithm place process of 412 KB, 312 KB, 112 KB and 510 KB.

UNIT - IV

- 5 Compare and contrast free space management and swap space management.

OR

What is disk structure ? Explain with diagram.

UNIT - V

- 6 What is file system ? Explain directory structure in detail.

OR

What is Authentication ? Differentiate between program threats and system threats.

SECTION - C

- 7 Consider the following set of process, with the length of CPU burst given in milliseconds.

Process	Bursttime	Priority
P ₁	10	3
P ₂	1	1
P ₃	2	3
P ₄	1	4
P ₅	5	2

The processes are assumed to have arrived in order P₁, P₂, P₃, P₄, P₅ and at time 0.

- (a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms FCFS, SJF, nonpreemptive priority (a smaller priority number implies a higher priority) and Round Robin (RR) (the quantum = 1).
- (b) What is the turn around time of each process mention in part (a) ?
- (c) What is the waiting time of each process mention in part a?

8 Explain the Banker's Algorithm with suitable examples.

OR

What is Deadlock ? Explain four necessary conditions for deadlock to occur with suitable examples. Describe the different methods for prevention and avoidance of dead lock.

- 9 Consider the following reference string :
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
Find the number of page fault with FIFO, OPTIMAL and LRV page replacement with four free frames which are initially empty. Which algorithm gives the minimum number of page faults ?

- 10 Suppose disk queue for the following blocks is created at a time.

50, 91, 150, 42, 130, 18, 140, 70, 60

Assuming the disk head initially at block no. 30 explain the disk scheduling for the following :

- (i) FCFS
- (ii) SSTF
- (iii) SCAN
- (iv) C-SCAN

What is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests for each.

- 11 Explain the Access Matrix Model. Briefly discuss the good of protection and domain of protection.